UNIVERSITY OF THESSALY
SCHOOL OF ENGINEERING

DEPARTMENT OF ELECTRICAL
& COMPUTER ENGINEERING

Νευρο-Ασαφής Υπολογιστική (Neuro-Fuzzy Computing)
Fall Semester 2024-2025
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Problem-set 2nd: **Teamwork** (2-persons)

Announcement: Friday, December 13, 2024
Delivery deadline: Sunday, January 12, 2025

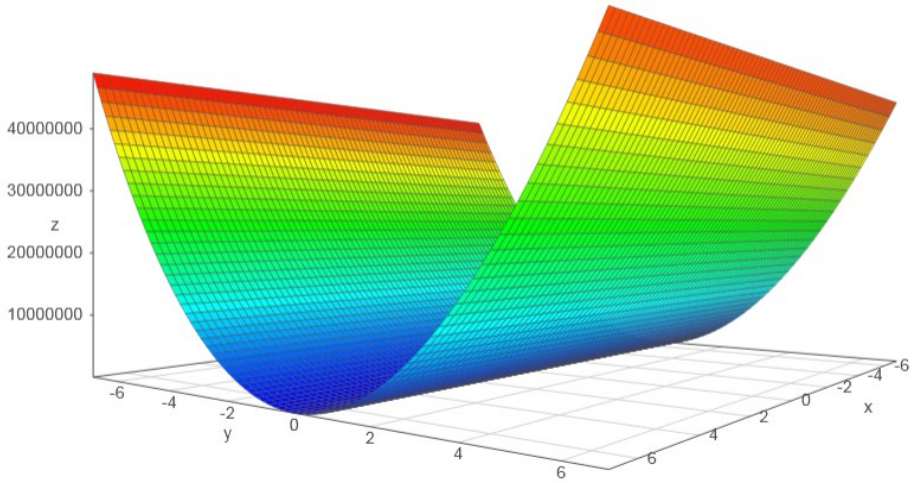
SECTION 1: Working with second-order minimizers: Conjugate Gradient and Newton



Problem-01

Find the minimum of the two-dimensional function:

$$F(\mathbf{w}) = w_1^2 + 10^6 w_2^2$$



with the Conjugate Gradient (Fletcher-Reeves), with initial guess $w(0)= [3, 3]^T$. Perform three iterations (if not finding the minimum earlier). Observe and comment on the slow/fast convergence towards the minimum $\mathbf{w}=(0, 0)^T$. Show your analytic calculations.



Problem-02

Find the minimum of the previous function using the Newton method and fixed step $\lambda_k=1$ (even though this is not a quadratic form).

SECTION 2: Working with model reduction techniques in MLPs



Problem-03

In the setting with $S^1= 10$ and learning rate $\alpha=0.05$ of **Problem-10** in the previous Problem-Set, apply the dropout technique (https://jmlr.org/papers/v15/srivastava14a.html) with dropout probability θ of hidden-layer neurons equal to $\theta=0.07$, then with $\theta=0.17$, and then with $\theta=0.3$. Apply the dropout during the training phase only, and only on the hidden layer units (not of course to the input neuron). Discuss the convergence

properties of the algorithm, as well as its accuracy, and contrast your findings with those of the previous exercise. Perform any additional experiments to figure out the operation of dropout as a generalization technique.



Problem-04

Along the lines of **Problem-10** of the previous Problem-Set, implement a MLP with S^1= 20. After training the neural network to convergence, prune a percentage (1% till 9%, step 2%) of the smallest – in magnitude – weights. Test the response (the accuracy) of the pruned MLP with respect to the unpruned version of it. Discuss the obtained results. Perform any additional experiments to figure out the operation of weight pruning as a generalization technique. Repeat the previous task, but prune now the same percentages of the largest – in magnitude – weights. Discuss again the obtained results.

SECTION 3: Working with modern optimizers



Problem-05

With respect to the function described in **Problem-01**, starting from a random point within the cube [-30,-40]x[-30,-40]:

1. Find the minimum of the function implementing by your own the AdaGrad optimizer (instead of gradient descent) with learning rate α=0.2. Plot the algorithm's trajectory on a contour plot of $F(\mathbf{x})$.
2. Change the learning rate to α=0.02, and repeat the same task.
3. Try out AdaGrad for the same objective function rotated by 45degrees, i.e., $F(w)=(w_1-w_2)^2+10^6(w_1+w_2)^2$. Does it behave differently?



Problem-06

With respect to **Problem-05**, solve [Bullet.1], but do not maintain a constant learning rate $\alpha(0)$=0.2, but implement a time-varying version of it as follows:

$$\alpha(t)=\frac{\alpha(0)}{\left(1+t\dfrac{a(0)}{n_{epochs}}\right)},$$

where $\alpha(0)$ is the initial learning rate, t the iteration number (epoch) and n_{epochs} the total number of epochs. Specifying the ratio $\alpha(0)/n_{epochs}$ decreases the learning rate from the previous epoch by the set amount. Plot the algorithm's trajectory on a contour plot and contrast the results with that obtained in the previous **Problem**.

SECTION 4: Working with Convolutional Neural Networks



Problem-07

Consider the following input image:

5	5	2	5	5
5	5	2	5	5
7	7	5	7	7
5	5	2	5	5
5	5	2	5	5

Let's assume we have the following 3x3 filters:

1	0	-1
2	0	-2
1	0	-1

1	2	1
0	0	0
-1	-2	-1

0	1	0
1	-4	1
0	1	0

A. Apply the given filters (by cross-correlation) to the above dataset. Use 'valid' padding and a stride of one.
B. Look at the structure of the filters. What do they do?



Problem-08

Let X be an input image and K be a filter. For all (x, y) ϵ Z^2 we define $T_{x,y}$ to be the translation operator that moves every point $X_{i,j}$ in the image by x in the x-direction and by y in the y-direction, i.e.: $T_{x,y}X_{i,j} = X_{i-x,j-y}$.
Show that the convolution operator is translation-equivariant, i.e., commutes with translations: $T_{x,y}X * K = T_{x,y}(X * K)$.



Problem-09

Max-pooling can be accomplished using ReLU operations.
A. Express max(a, b) by using only ReLU operations.
B. Use this to implement max-pooling by means of convolutions and ReLU layers.
C. How many channels and layers do you need for a 2x2 convolution?



Problem-10

Consider a convolutional neural network that is used to classify images into two classes. The structure of the network is as follows:

- INPUT: 100x100 grayscale images.
- LAYER-1: Convolutional layer with 50 5x5 convolutional filters.
- LAYER-2: Convolutional layer with 300 5x5 convolutional filters.
- LAYER-3: A max pooling layer that down-samples LAYER-2 by a factor of 4 (from 100x100 → 50x50)
- LAYER-4: Dense layer with 200 units
- LAYER-5: Dense layer with 300 units
- LAYER-6: Single output unit

How many weights does this network have?



Problem-11



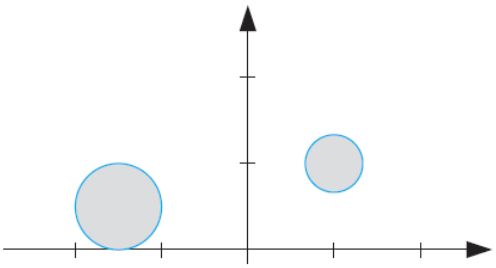
Your task is to implement fast convolutions with a kxk kernel. One of the algorithm candidates is to scan horizontally across the source, reading a k-wide strip and computing the 1-wide output strip one value at a time. The alternative is to read a k + Δ wide strip and compute a Δ-wide output strip.
A. Why is the latter preferable? Is there a limit to how large you should choose Δ?
B. Create a sample image of 228x228 and a series of 3x3, 7x7 and 11x11 kernels, and measure the execution time of the two alternatives. Comment on your results.

SECTION 4: Working with Radial Basis Function Neural Networks



Problem-12

Design an RBF network to perform the classification illustrated in the following figure. The network should produce a positive output whenever the input vector is in the shaded region and a negative output otherwise.





Problem-13

Write your own code to implement the gradient descent algorithm for the 1-S^1-1 RBF network. Train the network to approximate the function:

$$g(p) = 1 + \sin(p\pi/3) \text{ for } -3 \le p \le 3.$$

A. Select 20 data points at random from the interval $-3 \le p \le 3$.

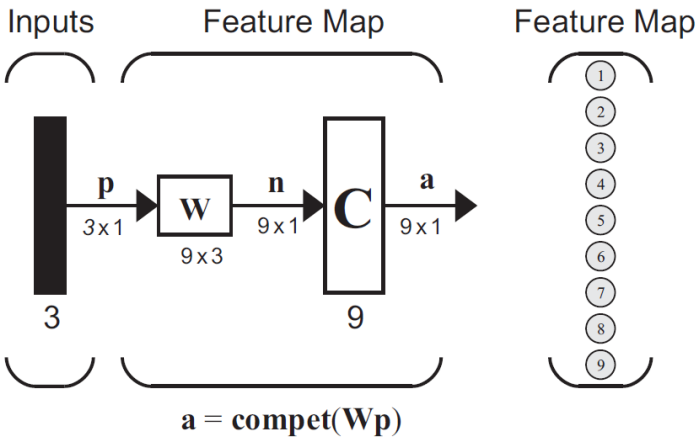
B. Initialize all parameters (weights and biases in both layers) as small random numbers, and then train the network to convergence. (Experiment with the learning rate α, to determine a stable value.) Plot the network response for $-3 \le p \le 3$, and show the training points on the same plot. Use 4, 8 and 10 centers. Try different sets of initial weights.

SECTION 5: Competitive Neural Networks



Problem-14

Consider the following competitive neural network and the respective one-dimensional feature map.



If the initial weight matrix is the following:

$$\mathbf{W} = \begin{bmatrix} 0.41 & 0.45 & 0.41 & 0 & 0 & 0 & -0.41 & -0.45 & -0.41 \\ 0.41 & 0 & -0.41 & 0.45 & 0 & -0.45 & 0.41 & 0 & -0.41 \\ 0.82 & 0.89 & 0.82 & 0.89 & 1 & 0.89 & 0.82 & 0.89 & 0.82 \end{bmatrix}^T$$

Train the network on the vector below for one iteration with learning rate α=0.1, and a neighborhood of radius equal to 1.

$$\mathbf{p} = \begin{bmatrix} 0.67 \\ 0.07 \\ 0.74 \end{bmatrix}.$$

Useful information:
The deadline is strict. It is possible to get an extension (up to 4 days), but you need to get the approval of the instructor, and this is going to cost a 10% penalty to the final grade of this Problem-Set. Delivery of the solutions' pdf (typeset or very very easy to read handwritten) is done by email to dkatsar@uth.gr. The subject of the message should be **strictly**: CE418-Problem set 02: AEM1-AEM2

Symbol interpretation:



It requires "algorithmic" and/or mathematical thinking.



It requires the development of code (in any language/platform you wish). The final deliverable should contain: a) The solution of the exercise, and b) The implementation source code.